INCH-POUND
MIL-DTL-32088C(OS)
19 April 2022
SUPERSEDING
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20 June 2011

DETAIL SPECIFICATION

INFLATION DEVICE, AUTOMATIC, FLU-12/P

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification provides the requirements for the manufacture, assembly, and packaging of the FLU-12/P automatic inflation device and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

DOD-STD-2101 - Classification of Characteristics

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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NAVAL AIR SYSTEMS COMMAND DRAWINGS

30003-849AS222 - Inflation Device, Automatic, FLU-12/P

(Copies of this document are available online via Joint Engineering Data Management Information and Control System (JEDMICS) at https://webjedmics.navair.navy.mil/webjedmics/index.jsp. Refer questions, inquiries, or problems to nani_customerservice@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General. The FLU-12/P inflation device shall be manufactured in accordance with 30003-849AS222. Lot numbers shall be in accordance with MIL-STD-1168.

3.3 Product baseline. A product baseline list shall be as specified (see 6.2).

3.4 Primary components. For the purposes of this specification, the explosive component is defined as a primary component (see 4.3.1). Primers shall have been manufactured within 24 months of the inflator manufacturing date. Prior to installation in the inflation devices, the explosive component shall meet the following requirements.

3.4.1 Bridgewire resistance. The bridgewire resistance of each explosive component shall be tested in accordance with 4.7.1. The bridgewire resistance value shall be in accordance with 30003-849AS222 or the acceptance test procedure approved by the Technical Authority (see 6.2 and 6.7.3).

3.4.2 Radiographic requirement. The explosive components shall be free of imperfections when examined in accordance with 4.7.2.

3.4.3 Ballistic requirement. The explosive components shall have been ballistically tested and accepted in accordance with the acceptance test procedure as approved by the Technical Authority (see 6.2).

3.5 Battery. Batteries shall have been manufactured within 48 months of the delivery date. Batteries installed in inflation devices shall have a minimum voltage of 12.0 volts when measured in accordance with 4.7.4.

3.6 Elastomeric materials. Elastomeric materials shall have been manufactured within 12 months of installation in the inflation device.

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3.7 Leakage at 40 pounds per square inch gauge (psig). The inflation devices shall not show any evidence of leakage when tested in accordance with 4.7.6.

3.8 Manual operation.

3.8.1 Lever operation without CO_2 cylinder. Each inflation device shall fully stroke the piercing pin. Sample inflation devices shall fully stroke the piercing pin when tested in accordance with 4.7.7.1.

3.8.2 Lever operation with CO_2 cylinder. Sample inflation devices shall fully stroke the piercing pin, perforate the CO_2 cylinder, and release the CO_2 gas when tested in accordance with 4.7.7.2.

3.9 Bleed circuit resistance. The bleed circuit resistance of each inflation device shall be accepted in accordance with the acceptance test procedures (see 6.2). Any changes in the procedure shall be reviewed and approved by the Technical Authority.

3.10 Visual examination requirement. Each inflation device shall conform to 30003-849AS222. Each inflation device shall be free of external damage; defects such as cracks, dents, corrosion, illegible or incomplete markings; missing or misassembled parts when inspected in accordance with 4.6.

3.11 Radiographic requirement. The inflation devices shall be free of imperfections when examined in accordance with 4.7.3.

3.12 Sensor circuit function. Each sensor assembly circuitry shall be electrically tested and accepted in accordance with the acceptance test procedures (see 6.2). Any changes in the procedure shall be reviewed and approved by the Technical Authority.

3.13 Environmental requirements.

3.13.1 Temperature and humidity (T&H) cycling. The inflation devices with batteries installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.1.

3.13.2 Vibration. The inflation devices with batteries and cylinder removed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.2.

3.13.3 Rain. The inflation devices with battery installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.3.

3.13.4 Salt fog. When tested in accordance with 4.7.5.4, the inflation devices with battery installed shall not function; the assembly shall not show evidence of corrosion, nicks, cracks, loose particles, or other defects; or otherwise be adversely affected.

3.13.5 Extended temperature. The inflation devices with battery installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.5 and 4.7.5.6.

3.13.6 Hydrostatic pressure. The inflation devices with battery installed shall not be adversely affected when tested in accordance with 4.7.5.7.

3.13.7 Differential pressure. The inflation devices with battery installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.8.

3.13.8 No-fire sensitivity. The inflation devices with battery installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.9.

3.13.9 Impact. The inflation devices with battery installed shall not function or otherwise be adversely affected when tested in accordance with 4.7.5.10.

3.14 Automatic operation. The inflation devices shall meet the following requirements when tested in accordance with 4.7.9.

3.14.1 Function time (see 6.7.1). The inflation devices shall function within 3 seconds after immersion in water in accordance with 4.7.9.1.

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3.14.2 Gas release. The CO_2 gas shall be fully released. The CO_2 gas shall not pass through any part of the inflation device other than the gas passages in the outlet port in accordance with 4.7.9.2.

3.14.3 Piercing pin. The piercing pin of the FLU-12/P shall fully stroke and remain in the fully stroked position after automatic inflation in accordance with 4.7.9.3.

3.14.4 Integrity. No part of the inflation device shall rupture or fragment in accordance with 4.7.9.4.

3.15 Workmanship. The inflation devices shall be constructed and finished in a manner to ensure compliance with all requirements of this specification and associated drawings.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 19 FLU-12/P inflation devices and 19 batteries shall be prepared (see 6.3). This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to starting production, is capable of producing items that comply with the technical requirements of the contract. Fourteen of these inflation devices shall be expended in the tests listed in table I and five shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample that has met the requirements of table II. First article sample acceptance shall consist of the tests shown in table I in the order specified.

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TABLE I. First article inspection.

Test	Requirement	Procedure	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Visual	3.10 and 3.15	4.6	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Radiographic	3.11	4.7.3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Battery voltage	3.5	4.7.4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
T&H	3.13.1	4.7.5.1	X																		
Vibration	3.13.2	4.7.5.2		X								X	X								
Rain	3.13.3	4.7.5.3			X																
Salt fog	3.13.4	4.7.5.4				X															
High temperature	3.13.5	4.7.5.5					X														
Low temperature	3.13.5	4.7.5.6						X						X							
Hydrostatic pressure	3.13.6	4.7.5.7							X												
Differential pressure	3.13.7	4.7.5.8								X											
No-fire sensitivity	3.13.8	4.7.5.9	X		X				X			X			X	X					
Impact	3.13.9	4.7.5.10									X										
Visual	3.10 and 3.15	4.6	X	X	X	X			X	X	X	X	X	X	X	X					
Radiographic	3.11	4.7.3	X	X	X	X			X	X	X	X	X	X	X	X					
Battery voltage	3.5	4.7.4	X	X	X	X			X	X	X	X	X	X	X	X					
Function at -20 °F	3.14	4.7.9.1						X								X					
Function at 70 °F	3.14	4.7.9.1	X	X	X	X			X	X	X	X	X	X							
Function at 120 °F	3.14	4.7.9.1					X								X						
Piercing pin lockout	3.14.3	4.7.9.3	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
Retained sample	---	4.4															X	X	X	X	X

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4.3 Conformance inspection. Conformance inspection shall consist of a lot inspection of all inflation devices as specified in table II and the sample lot acceptance tests as specified in table III. Any inflation device that does not meet the requirements of each test and examination in table II shall be removed from the production lot. Unless otherwise specified (see 6.2), the lot acceptance test sample size shall be in accordance with ANSI/ASQ Z1.4, General Inspection Level II. Inspection lot definition and formation shall be in accordance with ANSI/ASQ Z1.4. The lot size shall be between 26 and 1200 inflation devices. Lot acceptance inflation devices shall be subjected to the tests of table III in the order specified. Any failure not attributed to an error in test setup, procedure, or execution shall place the lot in a reject status (see 6.4).

TABLE II. Conformance inspection – lot inspections.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101) 1/, 2/
Primary component	Primer age	3.4	4.3.1	(M101)
Primary component	Single lot verification	3.4	4.3.1	(M102)
Primary component	Bridgewire resistance	3.4.1	4.7.1	(M103)
Primary component	Radiographic inspection	3.4.2	4.7.2	(M104)
Primary component	Ballistic acceptance	3.4.3	6.2	(C1)
Elastomeric materials	Age	3.6	4.3.1	(M105)
Sensor circuit	Function	3.12	---	(C2)
Inflation device	Leakage	3.7	4.7.6	(M106)
Inflation device	Bleed circuit resistance	3.9	4.7.8	(M109)
Inflation device	Radiographic inspection	3.11	4.7.3	(M110)

NOTES:

1/ (M) classification refers to Major characteristics (see 4.5).

2/ (C) classification refers to Critical characteristics (see 4.5).

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TABLE III. Conformance inspection – sample lot acceptance tests.

Item	Nature of test	Requirement	Test method	Quantity	Classification (DOD-STD-2101) 1/, 2/
Inflation device	Radiographic inspection	3.11	4.7.3	All	(M110)
Inflation device	Visual examination	3.10 and 3.15	4.6	All	(M111)
Inflation device	Battery voltage	3.5	4.7.4	All	(M112)
Inflation device	No-fire sensitivity	3.13.8	4.7.5.9	All	(M113)
Inflation device	Battery voltage	3.5	4.7.4	All	(M112)
Inflation device	Manual operation w/o cylinder	3.8.1	4.7.7.1	All	(M107)
Inflation device	Manual operation with cylinder	3.8.2	4.7.7.2	All	(M108)
Inflation device	Automatic operation	3.14	4.7.9	All	(C3)
Piercing pin	Lockout verification	3.14.3	4.7.9.3	All	(M114)

NOTES:

1/ (M) classification refers to Major characteristics (see 4.5).

2/ (C) classification refers to Critical characteristics (see 4.5).

4.3.1 Production lot. Lot numbers shall be in accordance with MIL-STD-1168. Inflation devices required for all test purposes shall be randomly selected from the production lot. Only primary components from a single lot shall be used in a production lot of inflation devices. One primary component production lot may be used in more than one inflation device production lot. The primary component lot number and age and the elastomer age shall be verified by visual inspection of the manufacture date. The results of the inspections listed in table II and tests listed in table III shall determine lot acceptance.

4.4 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with ANSI/ASQ Z1.4, General Inspection Level II. Two inflation devices in addition to the sample quantity shall be retained for investigative purposes. A change in inspection level shall be approved by the Technical Authority.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major, or Minor in accordance with DOD-STD-2101. Tests and examinations that verify Critical characteristics are identified by the symbol (C) and Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations that are not annotated with a classification code are classified as Minor.

4.6 Visual examination. Inflation devices shall be visually examined and meet the requirements of 3.10 and 3.15.

4.7 Tests.

4.7.1 Bridgewire resistance. The bridgewire resistance shall be measured with a test circuit that limits the bridge current to a maximum of 5 milliamperes. An instrument that is known to be accurate within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is met. Any explosive component not meeting the requirements of 3.4.1 shall be rejected.

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4.7.2 Explosive component radiographic examination. Each explosive component shall be radiographically examined in accordance with ASTM E1742/E1742M; any observable imperfections in assembly shall be cause for rejection. The explosive components shall be positioned on their sides for the most revealing exposure with the plane of the pins perpendicular to the x-ray source. All explosive components shall be identified utilizing a unique numbering system prior to examination. The explosive components shall be arranged sequentially on tray boards utilizing this system. Each radiographic plate shall carry a permanent means enabling identification of individual explosive components displayed thereon. The radiograph identification shall include the complete lot number and the span of numbers displayed. Discontinuities in the numbering sequence shall be identified on the x-ray. Defective explosive components found by the vendor shall be clearly identified on the x-ray and those defectives removed from the production lot.

4.7.3 Radiographic examination. Each inflation device shall be radiographically examined in accordance with ASTM E1742/E1742M; any observable imperfections in assembly shall be cause for rejection. The inflation devices shall be positioned on their sides for the most revealing exposure with the plane of the lever perpendicular to the x-ray source. All inflation devices shall be identified with serial numbers prior to examination. The inflation device serial numbers shall be in consecutive order beginning with the lowest number in each production lot. The inflation devices shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the inflation devices displayed thereon. The radiograph identification shall include the complete lot number and serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray. Defective inflation devices found by the vendor shall be clearly identified on the x-ray and those defectives removed from the lot.

4.7.4 Battery voltage. Battery age shall be verified by inspection of the manufacturer's manufacture date. Batteries shall be installed in inflation devices for the battery voltage test. The voltage of each inflation device shall be tested in accordance with the acceptance test procedures. Any changes in the procedure shall be reviewed and approved by the Technical Authority. Batteries that do not meet the requirements of 3.5 shall be replaced.

4.7.5 Environmental tests.

4.7.5.1 T&H cycling. The fully assembled inflation devices, including batteries and CO_2 cylinders, shall be supported on screen trays so that all areas are exposed to the prescribed atmospheric conditions at all times. The D port in the inflation devices shall be plugged to prevent intrusion of moisture into the devices. The T&H cycling scheduling shall be as specified in table IV.

TABLE IV. T&H cycling scheduling.

Day	Time	Cycling action
Monday	0800	Place test items in a chamber maintained at +70 °F at 50 percent relative humidity (RH).
	1200	Raise chamber temperature to +120 °F and the RH to 95 percent. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F.
Tuesday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from above chamber and immediately place in chamber maintained at -20 °F.
	1600	Remove test items from above and immediately place in a chamber maintained at 120 °F at 95 percent RH.

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TABLE IV. T&H cycling scheduling – Continued.

Wednesday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 0900.
	1200	Raise chamber temperature to +120 °F at 95 percent RH. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F.
Thursday	0800	Remove test items from above chamber and immediately place in a chamber maintained at +70 °F at 50 percent RH.
	1200	Remove test items from the above chamber and immediately place in a chamber maintained at -20 °F.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at +120 °F at 95 percent RH.
Friday	0800	Reduce chamber temperature to +70 °F at 50 percent RH. The chamber temperature shall reach +70 °F at 50 percent RH not later than 1300.
	1200	Raise chamber temperature to +120 °F at 95 percent RH. The chamber temperature shall reach +120 °F at 95 percent RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -20 °F.

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +120 °F at 95 percent RH. Evidence of cracks, deformation, distortion, function, or any other flaws shall be cause for rejection.

4.7.5.2 Transportation vibration. Vibration of the inflation device shall be equally divided at -20 °F, 70 °F, and 120 °F. The inflation device without battery or CO_2 cylinder shall be mounted on a fixture that allows vibration in three mutually perpendicular axes. In each axis, a resonance survey shall be conducted and the resonance points noted (see 6.7.2). A vibration dwell shall be conducted for 30 minutes at each of the four most severe resonance points. The dwell shall be peaked in case the resonant point shifts. After resonant dwell, the inflation device shall be vibration cycled for the time shown in table V. The vibration cycling shall be sinusoidal vibration to the levels shown on figure 1. The frequency shall be logarithmically swept from 5 Hz to 2000 Hz and return to 5 Hz in 20 minutes. Evidence of cracks, deformation, distortion, function, or any other flaws shall be cause for rejection.

TABLE V. Vibration test schedule/axis.

Resonances	**0**	**1**	**2**	**3**	**4**
Total Dwell Time (minutes)	0	30	60	90	120
Total Cycling Time (minutes)	180	150	120	90	60

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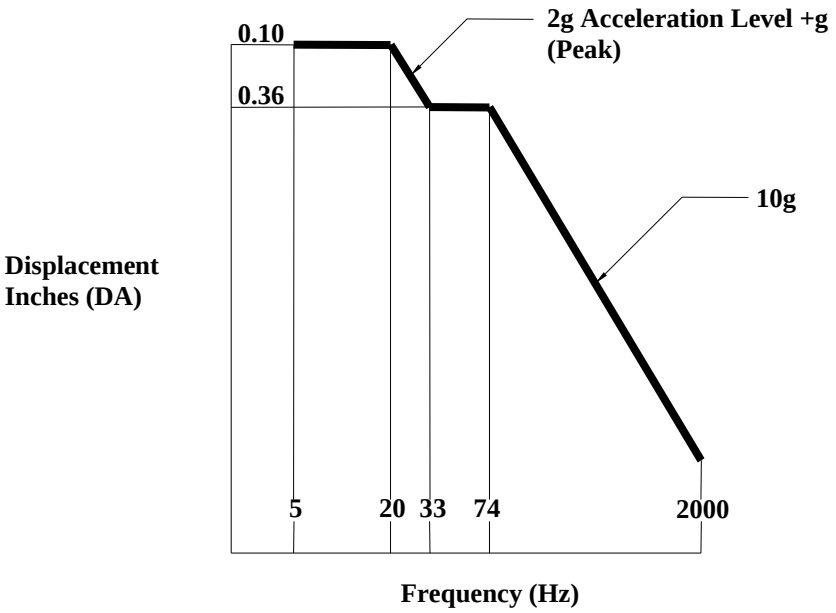


FIGURE 1. Vibration test levels.

4.7.5.3 Rain. The fully assembled inflation devices, including batteries and CO_2 cylinders, shall be placed on plastic screen trays to allow air to circulate freely, water to drain easily, and exposing all surfaces to the environment. The rainfall rate shall be adjusted to 4±1 inches per hour. The rain water conductivity shall be 350±35 micromho/centimeter. These conditions shall be maintained for 30 minutes. The tray shall then be rotated 90 degrees and the test repeated. The tray shall be rotated, and the test repeated until all four sides have been tested. The inflation devices shall then be dried at 120 °F for 24 hours. Evidence of function shall be cause for rejection.

4.7.5.4 Salt fog. The inflation assembly shall be subjected to the conditions of the salt fog test in accordance with ASTM B117. The test duration is 48 hours and then 48 hours drying. Evidence of corrosion, nicks, cracks, loose particles, function, or other defects shall be cause for rejection.

4.7.5.5 Extended high temperature. The inflation device shall be temperature conditioned at 160±5 °F for 48.5 hours. The inflation device shall then be removed from the conditioning chamber and allowed to stabilize at 120±3 °F. The inflation device shall then be immediately tested in accordance with 4.7.9.1. Evidence of cracks, flaking, deformation, distortion, splintering, or other defects shall be cause for rejection.

4.7.5.6 Extended low temperature. The inflation devices shall be temperature conditioned at 20±3 °F for 48.5 hours. The inflation devices shall then be removed from the conditioning chamber and immediately tested in accordance with 4.7.9.1. Evidence of cracks, flaking, deformation, distortion, splintering, or other defects shall be cause for rejection.

4.7.5.7 Hydrostatic pressure. A discharged carbon dioxide cylinder shall be inserted in the inflation assembly with a maximum torque of 20 inch-pounds. The assembly shall be subjected to a hydrostatic pressure of 1500±100 psig applied to the outlet of the assembly for a minimum of 5 minutes. Evidence of leakage or deformation of the body shall be cause for rejection.

4.7.5.8 Differential pressure. The outlet port of the inflation device shall be blocked for this test. The inflation device equipped with a filled carbon dioxide cylinder shall be subjected to three cycles of surge air pressure of 40±2 psig for a duration of 50±20 milliseconds. Evidence of damage or function shall be cause for rejection.

4.7.5.9 No-fire sensitivity. The inflation devices shall be temperature conditioned at 70±5 °F for a minimum of 4 hours. To simulate water conductivity, a 160-K-ohm±10 percent resistor shall be connected between the sensor housing contact and the body of the inflator for a minimum of 60 seconds.

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4.7.5.10 Impact. A CO_2 filled cylinder shall be fully installed in the inflation device hand tight. The inflation device shall be free fall dropped from a minimum height of 4 feet onto a fixed steel plate twice on each face. Evidence of cracks, deformation, distortion, function, or any other flaws shall be cause for rejection.

4.7.6 Leakage at 40 psig. The leakage test shall be performed on inflation devices without shear wire or batteries installed. The lever shall be actuated through one complete cycle and returned to its original position. A pressure of 40±2 psig shall be applied at the primer port of the inflation device for a minimum of 10 seconds. While the pressure is applied, the inflation device shall be observed for leakage. The test shall be performed on the entire lot. Any detectable pressure drop shall be cause for rejection of the inflation device.

4.7.7 Manual operation.

4.7.7.1 Lever operation without CO_2 cylinder. The lever shall not be secured to the body with shear wire. Move the lever through three cycles to minimize O-ring set. The inflation device shall be oriented vertically with the CO_2 cylinder inlet port up. The inflation device shall be rotated approximately 10 degrees in the direction of lever rotation. A 3±0.5-pound weight shall be attached to the free end of the lever. The lever shall operate through one normal cycle. The piercing pin shall advance to its maximum point of travel and fully retract. This test shall be applied to 4 percent of the lot quantity rounded to the nearest whole number. That quantity is to be taken from the lot acceptance test (LAT) samples. If that 4 percent exceeds the sample quantity, test only the LAT sample. The LAT sample will then be tested in accordance with paragraph 4.7.7.2.

4.7.7.2 Lever operation with CO_2 cylinder. A CO_2 filled cylinder shall be fully installed in the inflation device hand tight. The shear wire shall not be installed. The CO_2 cylinder shall then be removed and its diaphragm face examined for any evidence of puncture or indentations by the piercing pin. Upon completion of this inspection, the cylinder shall be reinstalled in the inflation device. The inflation device shall be oriented vertically with the CO_2 cylinder up. A 15±0.5-pound weight shall be attached to the free end of the lever. The piercing pin shall advance to its maximum point of travel, pierce the diaphragm of the CO_2 cylinder, and release the CO_2 gas. Any inflation device that does not meet the requirements of 3.8.2 shall be rejected. This test shall be applied to the units tested in accordance with 4.7.7.1.

4.7.8 Bleed circuit resistance. The bleed circuit resistance shall be tested in accordance with the acceptance test procedure. Any changes in the procedure or resistance value shall be reviewed and approved by the Technical Authority. If the resistance reading is not in accordance with the approved procedure, the inflation device shall be rejected.

4.7.9 Automatic operation tests.

4.7.9.1 Functional test. The inflation devices shall be equally divided for temperature conditioning at -20±5 °F, 70±5 °F, or 120±5 °F for a minimum of 4 hours. The inflation devices shall be functioned within 3 minutes after removal from temperature conditioning. The inflation devices shall be functioned by immersion in water at 32+5/-0 °F, and 100+50/-10 micromho/centimeter conductivity. The time from immersion in water to start of release of CO_2 gas shall be measured.

4.7.9.2 Gas release. The functioned inflation devices shall be inspected for complete release of gas from the CO_2 cylinder through the outlet port (see 3.14.2).

4.7.9.3 Piercing pin. Automatically actuated inflation devices shall be inspected for complete stroke of the piercing pin (see 3.14.3). To verify complete stroke and lockout of the piercing pin, the automatically actuated inflation devices shall be able to withstand a minimum axial force of 260 Newtons (58.4 pound-force) on the piercing pin without it being pushed back into the inflator bore. Verification can also be done by the installation of a second CO_2 cylinder without pushing the piercing pin back into the bore.

4.7.9.4 Integrity. The functioned inflation devices shall be inspected for integrity (see 3.14.4).

4.7.10 Acceptance criteria. The total number of allowed defective units in a first article sample or lot acceptance sample shall be zero. Any defects will reject the lot represented. Screening of defects as a corrective action and the method of screening shall be subject to approval by the Technical Authority.

4.7.11 Retest. There shall be no retests.

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4.7.12 Test failure. If a test failure is attributable to an assignable cause, excluding the inflation devices, the original individual test results should be discarded and the complete test repeated.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The FLU-12/P inflation device is designed to inflate life preservers either manually or automatically by opening a CO_2 cylinder and releasing the gas into the life preserver through an outlet. This inflation device was designed specifically for use in the Mk1 Mod 0 life preserver used by Navy deck crew personnel and has no commercial application. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority. Current stock of inflation devices may be used until depleted; future requisitions of the inflation device should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Designation of inflation device (FLU-12/P).

c. When first article inspection is required (see 3.1).

d. Requirements for a product baseline list (see 3.3).

e. Requirements for Government approval of the acceptance test procedures (see 3.4.1, 3.4.3, 3.9, and 3.12).

f. The lot acceptance test sample size if other than specified (see 4.3).

g. Production lot size and test activity (see 4.3.1 and 6.4.1).

h. Requirements for sampling if other than specified (see 4.4).

i. Packaging and marking requirements (see 5.1 and 6.6).

j. Items of data required for each first article and production lot (see 6.3).

k. That the safety precaution requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of NAVSEA OP 5, Ammunition and Explosives Ashore, are applicable.

6.3 First article inspection. The sample inflation devices are to be delivered to the activity designated by the contracting agency for test. Any quantity of production prior to notification by the contracting agency of first article sample acceptability will be at the contractor's risk. The first article sample inflation devices will not be applied as a part of the quantity specified for delivery by the contract.

6.4 Conformance inspection. If a subsequent failure investigation identifies a root cause and corrective action, the lot can be accepted via variance if defective units can be identified and removed from the lot. If the failure is attributed to a test anomaly caused by human error or equipment malfunction, a unit retained for investigative purposes can be used to continue testing with the approval of the Technical Authority. The reject status can only be removed by the Technical Authority.

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6.4.1 Production lot. Inflation devices required for all test purposes will not be applied as part of the quantity specified for delivery by the contract. Inflation devices selected for the ballistic test are to be delivered to the activity designated by the contracting agency.

6.5 Data. For the information of contractors and contracting officers, any of the data specified in the subparagraphs below, applicable documents listed in section 2 of this specification, or referenced in lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.5.1 Standard deliverables. In addition to the delivery of inflation devices, the contract may require delivery of the following:

a. Radiographic plates of FLU-12/P inflators.

b. Radiographic review report.

c. Radiographic sample plates and techniques.

d. Requests for variance and engineering change proposal (ECP).

e. Progress and management report.

f. Test and inspection report if contractor tested.

g. Ammunition data cards.

6.6 Contract packaging and marking. The following packing and marking requirements should be specified in the contract.

6.6.1 Preservation and packing. Level A and commercial preservation/packing should be in accordance with 49 CFR 171-178. The contractor is responsible for performance-oriented packing certification and documentation in accordance with 49 CFR 171-178. Level A preservation and packing should be used for packaging of all production lots. Inflation devices should be packaged in an inner container in accordance with SPI-PHST-283. Inner containers should be packed in the outer container in accordance with SPI-PHST-283. For an explanation of terms, see 6.6.2.

6.6.2 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28, when applicable, is used to clarify terms.

6.6.3 Markings.

6.6.3.1 Special markings. Marking of exterior containers should be in accordance with 49 CFR 171-178. Markings should be applied by the contractor or their agent.

6.6.3.2 Normal markings. Unless otherwise specified (see 6.2), the marking information on unit packages and shipping containers should be as specified on SPI-PHST-283. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6.7 Definitions.

6.7.1 Function time. The time from immersion of the inflation device in water to start of release of CO_2 gas.

6.7.2 Resonance. The ratio of vibration output versus input of 2 or greater.

6.7.3 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8 Subject term (keyword) listing.

Bleed circuit

Bridgewire

Elastomeric material

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Inflator

Primary component

Sensor circuit

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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